Access Modifier:

In **TypeScript**, access modifiers are used to control **where a class property or method can be accessed**.

There are **3 main access modifiers**:

Modifier	Inside class	Child class	Outside class
public	✅	✅	✅
protected	✅	✅	❌
private	✅	❌	❌

1. public

Can be accessed **anywhere**. It is also the **default** modifier.

```
class Student {
   public name: string = "Arya";

   displayName() {
      console.log(this.name);
   }
}

let s = new Student();
console.log(s.name);  // ✅
```

You can even write:

```
name: string = "Arya";
```

which is effectively public.

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2. private

Can be accessed **only inside the same class**.

```
class Student {
   private age: number = 28;

   displayAge() {
      console.log(this.age);  // ✅
   }
}

let s = new Student();
```

```
console.log(s.age);  // ❌ Error
```

A child class also cannot directly access private.

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3. protected

Can be accessed **inside the class and its child/subclass**, but **not from outside**.

```
class Student {
   protected age: number = 28;
}

class CollegeStudent extends Student {
   displayAge() {
      console.log(this.age);  // ✅
   }
}

let s = new CollegeStudent();

console.log(s.age);  // ❌ Error
```

Easy way to remember

public → Everyone
protected → Class + Child class
private → Only same class

For interviews, you can define it as:

Access modifiers in TypeScript are keywords used to control the accessibility of class members such as properties and methods. The main access modifiers are public, protected, and private.